

Alzheimer's Disease Detection System

AI-Assisted MRI Analysis & Clinical Decision Support Report

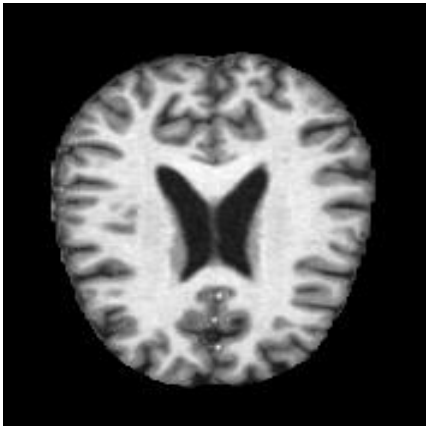
Report ID:	a18cb583-0150-42fe-acba-8e9979bf57f4	Date:	2026-07-31 06:36:11 UTC
Patient Name:	Barbara McClintock Kern	Patient ID:	OAS1_0405_MR1
Age:	85	Blood Group:	O+
Reported Symptoms:	Memory loss disrupting daily life, Decreased or poor judgment, Withdrawal from work or social activities, Comprehensive neuro-diagnostic workup requested.	Model Version:	custom_cnn

Predicted Disease Stage:	Non Demented	Confidence Score:	32.42%
--------------------------	--------------	-------------------	--------

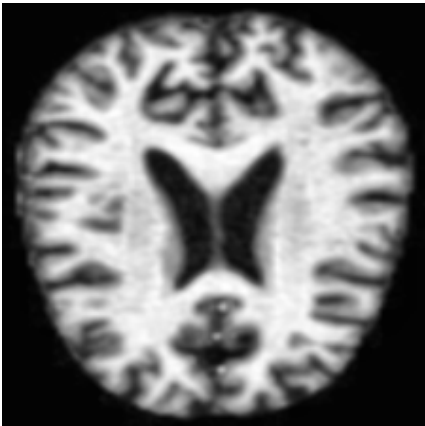
Medical Image Processing & Preprocessing Pipeline Summary

Image Quality Score:	91.8/100 (Excellent)	Processing Duration:	707.32 ms
Noise Reduction:	Applied (Gaussian - 8 ms)	Contrast Enhancement:	CLAHE Active (10 ms)
Intensity Normalization:	Min-Max [0.0, 1.0]	ROI Detection & Crop:	Brain Contour Crop (6 ms)

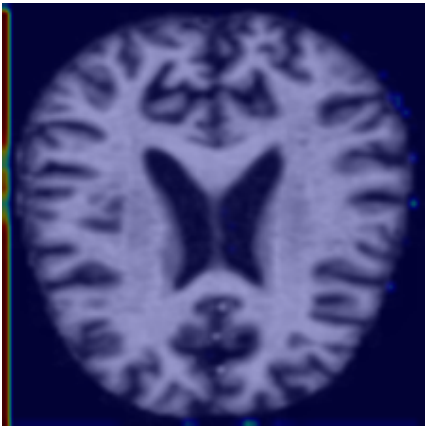
Visual MRI Comparison & Explainability Analysis



1. Original MRI Scan



2. Enhanced MRI



3. Grad-CAM Overlay

Class Probabilities Distribution

Disease Stage	Probability Score	Confidence Bar
Non Demented	32.42%	32%
Very Mild Demented	29.06%	29%
Mild Demented	24.06%	24%
Moderate Demented	14.46%	14%

IMPORTANT NOTICE: MEDICAL DISCLAIMER: This AI prediction and report are generated for research, educational, and decision-support purposes only. This software is NOT a certified diagnostic medical device. Results must be independently verified by a qualified medical professional.